

Biological Networks

Yongjin Park

02 March 2026

Why do we study biological networks?

- Genes and proteins function through **interactions** with each other
- Understanding interactions helps us:
 - Map cellular pathways and processes
 - Identify disease mechanisms
 - Predict gene function
 - Design targeted therapies

Why networks matter in cancer biology

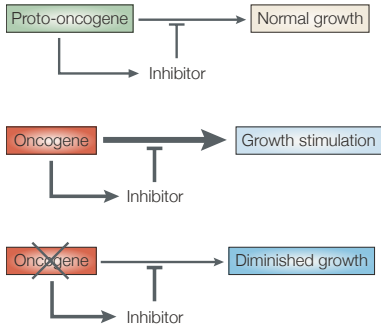
Challenge in cancer therapy:

- Tumour cells develop resistance to single-target drugs
- Genetic redundancy provides backup pathways
- Need to hit multiple targets simultaneously

Network solution:

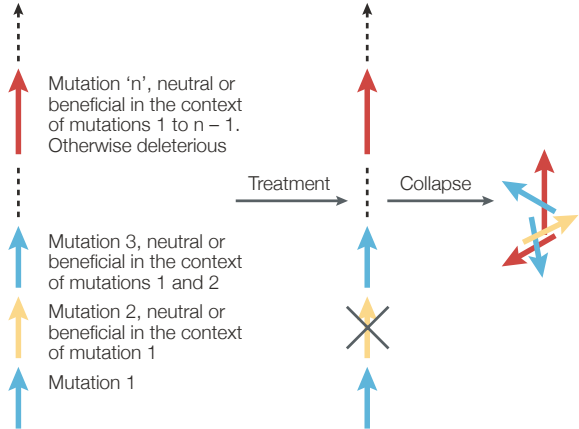
- Synthetic lethality: exploit genetic interactions
- Target pairs of genes that are lethal together
- Example: BRCA1/2 mutations + PARP inhibitors

Networks of oncogene and inhibition intertwined



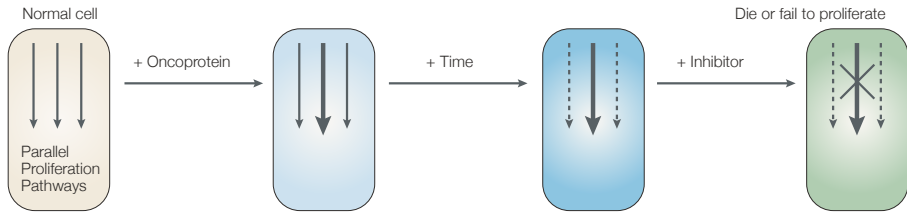
- Oncogenes may regulate other genes
- Which one should be suppressed?
- Which one can be?

Transformed



Normal

Cancer treatment may need to deal with multiple genes. . .



(Kaelin 2005)

Today's lecture

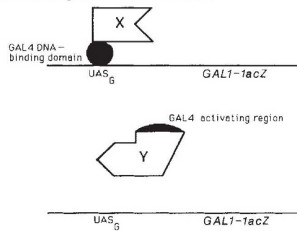
- 1 How do we measure protein/gene interaction networks?
- 2 How can we use network data?
- 3 New frontiers: cell-cell communication and regulatory perturbation

Y2H to capture protein-protein interactions

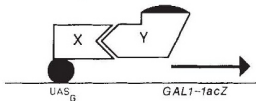
a
Native GAL4



b
Individual hybrids with GAL4 domains



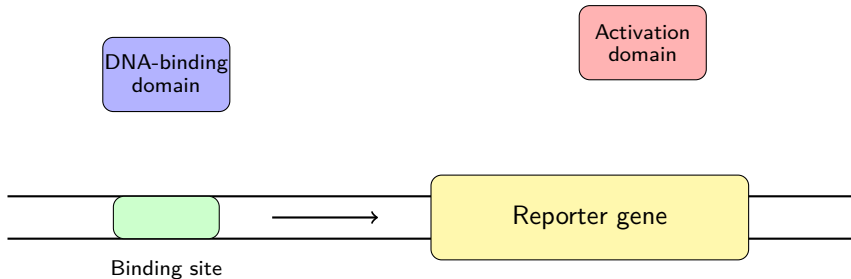
c
Interaction between hybrids reconstitutes GAL4 activity



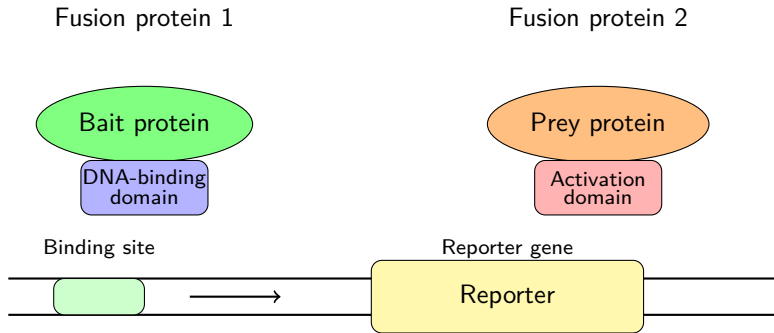
- Yeast Two Hybrid
- GAL4-lacZ reporter system
- Transcription factors have two separable domains: DNA-binding (BD) and activation (AD)
- Fuse BD to “bait” protein, AD to “prey” protein
- If they interact $\Rightarrow$ reconstitute functional TF $\Rightarrow$ reporter gene ON

(Fields and Song 1989)

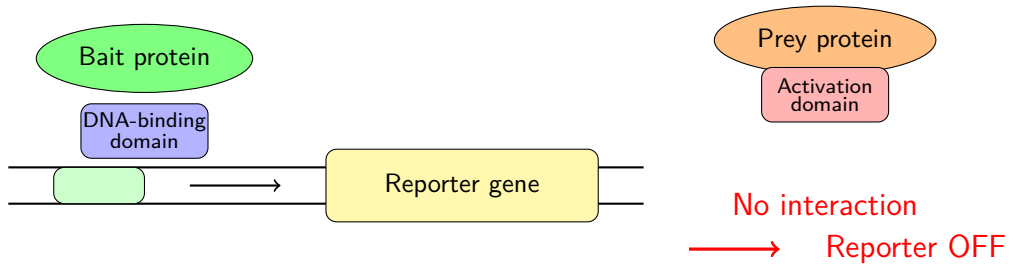
Synthesize TF binding sites with reporter genes



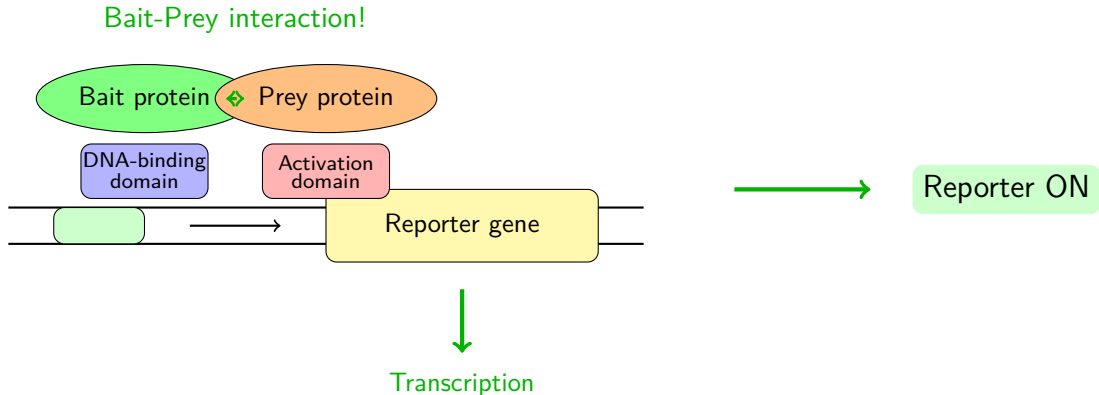
Create fusion proteins



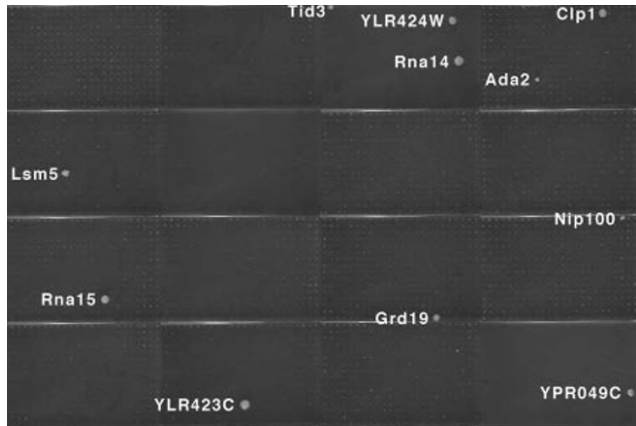
Negative: No interaction between bait and prey



Positive: Bait-prey interaction reconstitutes TF



First genome-wide Y2H screen

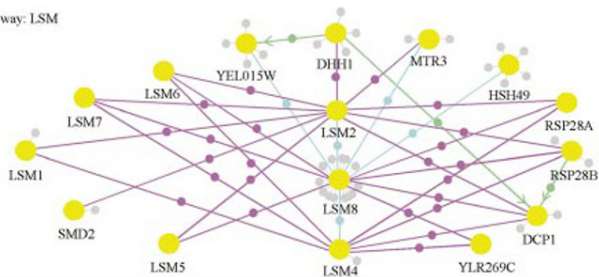


- Array of ~6000 prey proteins (AD fusions)
- Each bait (BD fusion) screened against entire array
- 192 bait proteins tested
- Growth on selective media = positive interaction

(Uetz et al. 2000)

First genome-wide Y2H screen

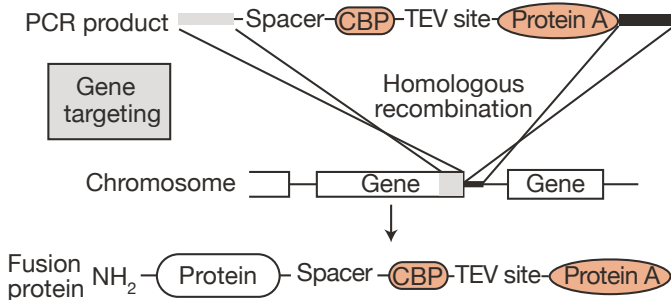
Pathway: LSM



- Array of ~6000 prey proteins (AD fusions)
- Each bait (BD fusion) screened against entire array
- 192 bait proteins tested
- Growth on selective media = positive interaction

(Uetz et al. 2000)

Affinity purification–mass spectrometry (AP-MS)

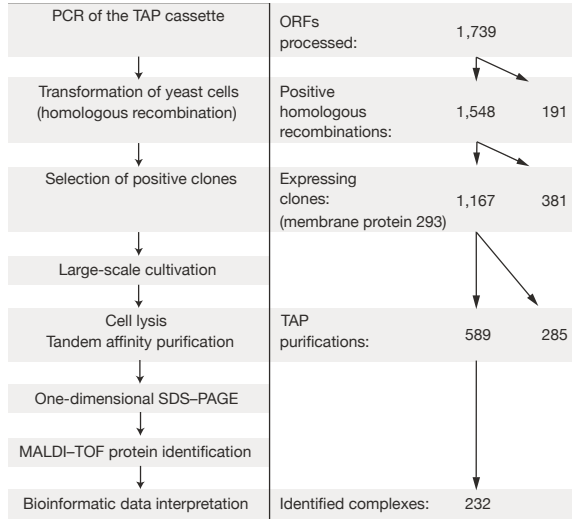


TAP tag (Tandem Affinity Purification):

- CBP (Calmodulin Binding Peptide) + Protein A (binds IgG).
- Two-step purification → reduce background

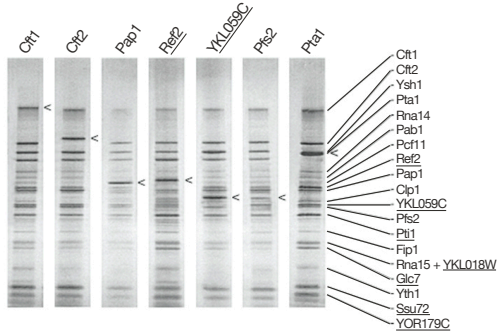
(Gavin et al. 2002)

AP-MS workflow (Gavin et al. 2002)

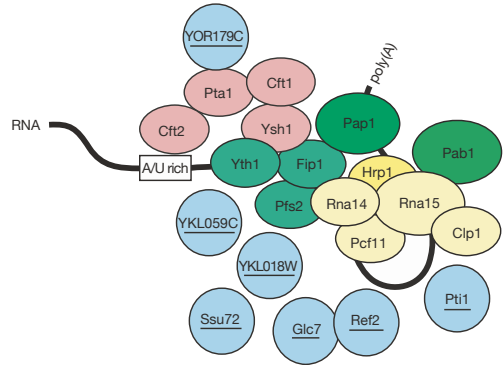


AP-MS network analysis

Purification



Mass Spec result:



(Gavin et al. 2002)

Y2H vs. AP-MS: True or False?

Question 1: Y2H detects only direct binary protein-protein interactions, while AP-MS can identify both direct and indirect interactions within protein complexes.

Y2H vs. AP-MS: True or False?

Question 1: Y2H detects only direct binary protein-protein interactions, while AP-MS can identify both direct and indirect interactions within protein complexes.

TRUE - Y2H requires direct physical contact between bait and prey to reconstitute the transcription factor. AP-MS co-purifies everything bound to the tagged protein, including indirect interactions mediated by other proteins in the complex.

Y2H vs. AP-MS: True or False?

Question 2: Y2H is an in vivo method (interactions measured in living cells), while AP-MS is an in vitro method (interactions measured after cell lysis).

Y2H vs. AP-MS: True or False?

Question 2: Y2H is an in vivo method (interactions measured in living cells), while AP-MS is an in vitro method (interactions measured after cell lysis).

TRUE - Y2H detects interactions in living yeast cells (in vivo, though in artificial nuclear context). AP-MS requires cell lysis before purification and mass spec analysis (in vitro).

AP-MS Challenge: True or False?

Question 3: In AP-MS, any protein detected by mass spectrometry is definitely part of the protein complex.

AP-MS Challenge: True or False?

Question 3: In AP-MS, any protein detected by mass spectrometry is definitely part of the protein complex.

FALSE - Need to distinguish true interactors from contaminants. Solutions: (1) Compare to negative controls, (2) Quantitative mass spec (abundance ratios), (3) Reciprocal tagging experiments, (4) Statistical scoring (e.g., hypergeometric p-value).

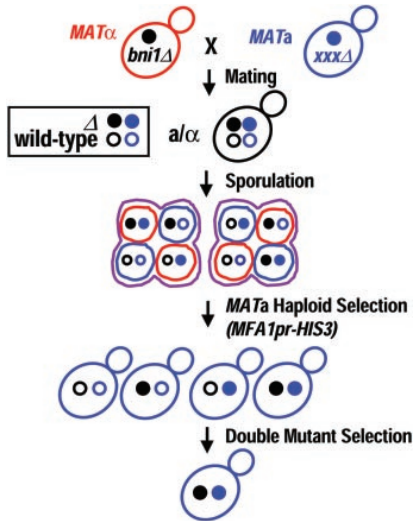
Genetic interaction networks — synthetic lethality

Systematic Genetic Analysis with Ordered Arrays of Yeast Deletion Mutants

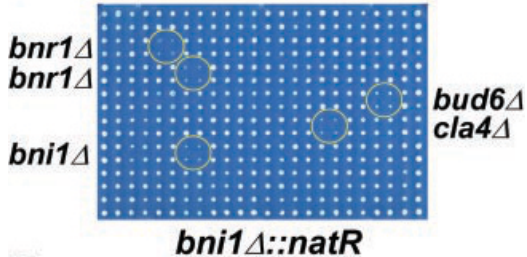
Amy Hin Yan Tong,^{1,2} Marie Evangelista,³ Ainslie B. Parsons,^{1,2}
Hong Xu,^{1,2} Gary D. Bader,^{4,5} Nicholas Pagé,⁶ Mark Robinson,¹
Sasan Raghizadeh,⁷ Christopher W. V. Hogue,^{4,5}
Howard Bussey,⁶ Brenda Andrews,^{2*} Mike Tyers,^{2,5*}
Charles Boone^{1,2,3*}

- Genetic interactions reveal functional relationships
- Most salient phenotype in yeast?
- How many cells within each strain?

(Tong et al. 2001)

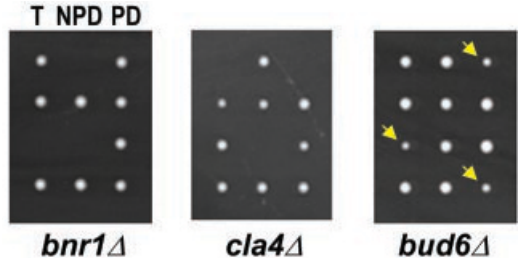
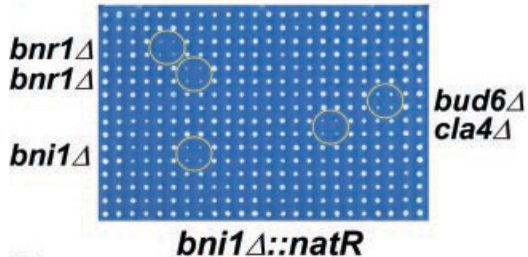


Synthetic Gene Array creates double mutants



(Tong et al. 2001)

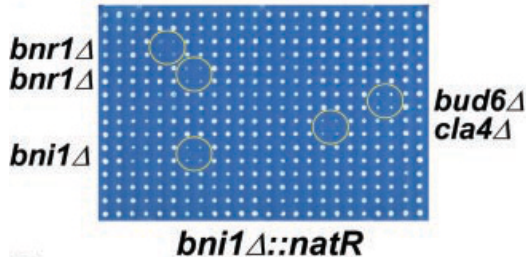
Synthetic Gene Array creates double mutants



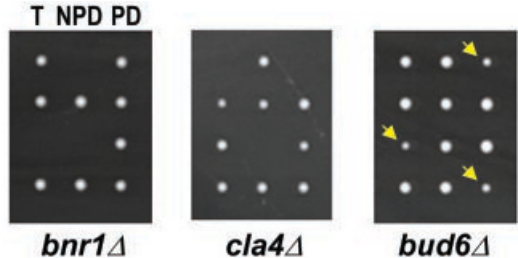
- Query mutation (*bni1Δ::natR*) crossed against deletion array.
- Smaller colonies = synthetic lethal/sick

(Tong et al. 2001)

Synthetic Gene Array creates double mutants



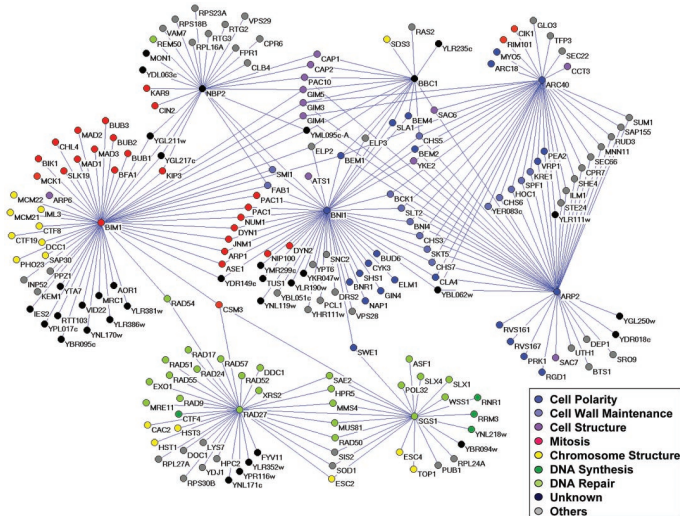
- Query mutation (*bni1Δ::natR*) crossed against deletion array.
- Smaller colonies = synthetic lethal/sick



- Tetrad analysis confirmation.
- T=tetrad, NPD=non-parental ditype, PD=parental ditype.
- Yellow arrows = double mutants with reduced growth.

(Tong et al. 2001)

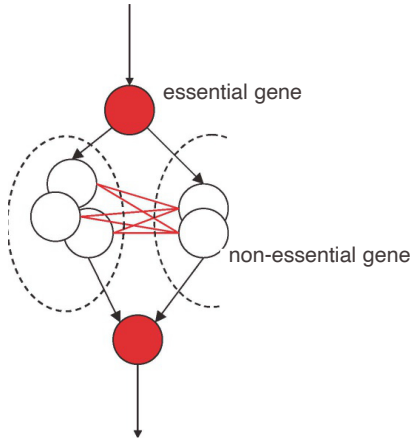
SGA reveals functional hub-and-spoke patterns



- Hub-and-spoke architecture
- Genes in same pathway cluster together
- Hubs (e.g., BNI1, ARP2, RAD27) are functionally critical
- Predict function of unknown genes by their neighbours

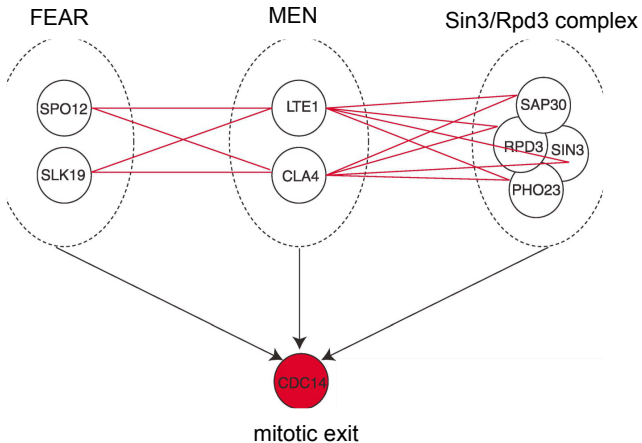
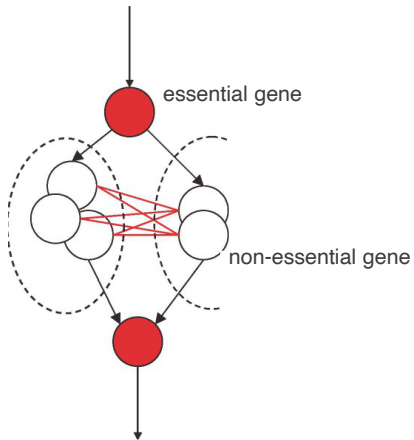
(Tong et al. 2001)

How should we think of synthetic lethality?



(Ye et al. 2005)

How should we think of synthetic lethality?



(Ye et al. 2005)

Today's lecture

- 1 How do we measure protein/gene interaction networks?
- 2 How can we use network data?
- 3 New frontiers: cell-cell communication and regulatory perturbation

Network is a data set

Graph $G = (V, E)$:

- **Vertices** (nodes): $V = \{v_1, v_2, \dots, v_n\}$ — genes/proteins
- **Edges**: $E = \{(v_i, v_j)\}$ — interactions between genes
- **Adjacency matrix**: $A_{ij} = 1$ if edge exists, 0 otherwise
- **Degree**: $d_v = \sum_u A_{uv}$ — number of neighbours

Example:

- BioGRID: $n \approx 18,000$ genes, $m \approx 500,000$ interactions
- Sparse: most genes connect to few others (not all-to-all)
- Scale-free: few hubs with many connections, most with few

We can use igraph in R

```
library(igraph)
```

- We can use igraph in R or Python or C++.

Let's take a look at some classic examples

Download some from Mark Newman's website:

```
#' @param .src www location
#' @param .dir downloading local directory
#' @param .zip zip file name
#' @param .tgt target gml file name
.read.online <- function(.src,
                        .dir="./data/network/",
                        .zip = str_c(.dir,basename(.src)),
                        .tgt = str_replace(.zip, "zip$", "gml")){
  if.needed(.tgt, {
    .gml <- basename(.tgt)           # remove the dir name
    download.file(.src, .zip)         # download from the web
    unzip(.zip, files=.gml, exdir = .dir) # unzip
  })
  return(igraph::read_graph(.tgt, format="gml")) # read them into memory
}
```

```
karate <- .read.online("http://www-personal.umich.edu/~mejn/netdata/karate.zip")
polblog <- .read.online("http://www-personal.umich.edu/~mejn/netdata/polblogs.zip")
netsci <- .read.online("http://www-personal.umich.edu/~mejn/netdata/netscience.zip")
football <- .read.online("http://www-personal.umich.edu/~mejn/netdata/football.zip")
condmat <- .read.online("http://www-personal.umich.edu/~mejn/netdata/cond-mat-2005.zip")
```

Handy functions for visualizing a network

We want to match graph vertices with the layout coordinates found by `igraph::layout_nicely` and make it easy to show in `ggplot` or something else.

```
##' @param G igraph object  
##' @param layout.file local file  
find.nice.layout <-  
  function(G, layout.file) {  
  
    if(!file.exists(layout.file)){  
      layout.dt <-  
        layout_nicely(G) %>%  
        as.data.table()  
      saveRDS(layout.dt, layout.file)  
    }  
    readRDS(layout.file)  
  }  
}
```

```
##' @param G igraph object  
##' @param .name file header  
apply.layout.dt <- function(G, .name){  
  .out.file <-  
    str_c(fig.dir,"/",.name,".rds")  
  
  .out <-  
    find.nice.layout(G, .out.file) %>%  
    dplyr::mutate(v = 1:dplyr::n()) # index  
  
  as_long_data_frame(G) %>%  
    as.data.table %>%  
    merge(.out, by.x= "from", by.y="v") %>%  
    merge(.out, by.x= "to", by.y="v",  
          suffixes=c(".from",".to"))  
}
```

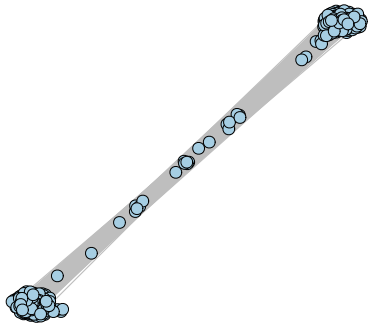
Plotting a graph with points and segments in ggplot

```
plot.layout.dt <- function(.dt, .palette="Paired", .palette.e = "Set2", .esize = .5, .vsize = 5, .add.lab=TRUE){
  if(!("k.from" %in% colnames(.dt))) .dt[, k.from := 1] # add membership
  if(!("k.to" %in% colnames(.dt))) .dt[, k.to := 1] # add membership
  .rename <- function(xx) { colnames(xx) <- c("v", "V1", "V2", "k"); xx } # Find a consolidated
  .dt.v <- # set of vertices
    rbind(.dt[, .(`from`, `V1.from`, `V2.from`, `k.from`)] %>% .rename(),
          .dt[, .(`to`, `V1.to`, `V2.to`, `k.to`)] %>% .rename()) %>%
    dplyr::mutate(k = as.factor(k)) %>% unique

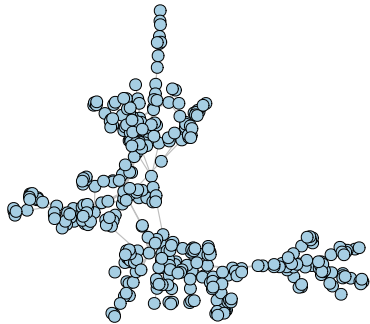
  .aes.e <- aes(V1.from, V2.from, xend=V1.to, yend=V2.to) # edge colours
  if("k.edge" %in% colnames(.dt)) {
    .aes.e <- aes(V1.from, V2.from, xend=V1.to, yend=V2.to, colour=as.factor(k.edge))
  }
  plt <- ggplot() + theme_void() # show no axis
  if("k.edge" %in% colnames(.dt)) {
    plt <- plt +
      geom_segment(.aes.e, data = .dt, size = .esize) + # draw edges
      scale_colour_brewer(palette=.palette.e, na.value="grey", guide="none")
  } else {
    plt <- plt +
      geom_segment(.aes.e, data = .dt, size = .esize, colour="gray") # draw edges
  }
  plt <- plt +
    geom_point(aes(V1, V2, fill=k), data = .dt.v, pch=21, stroke=.1, size = .vsize) + # vertices
    scale_fill_brewer(palette = .palette, guide="none") # vertex colouring

  if(.add.lab){ plt <- plt + geom_text(aes(V1, V2, label=v), data = .dt.v, size = 3) }
  return(plt)
}
```

Social network examples

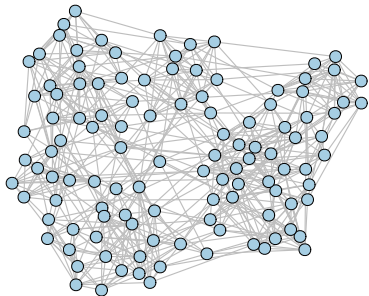


(Adamic and Glance 2005)



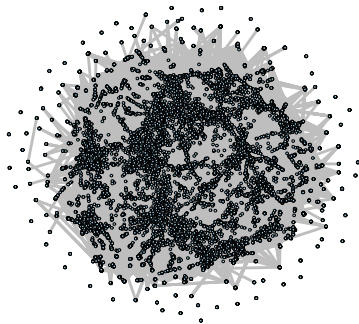
- Coauthorship network of scientists working on network theory and experiment
 - Showing only the largest component
- (Newman 2006)

More examples



- Network of American football games between Division IA colleges during regular season Fall 2000.

(Girvan and Newman 2002)



(Newman 2001)

- Showing densely-connected part (iterative degree cutoff ≥ 5)

Human Reference Protein-Protein interaction map

- Download the data from interactome-atlas.org

```
db.url <- "http://www.interactome-atlas.org"
ppi.net.file <- "../data/HI-union.tsv.gz"
if.needed(ppi.net.file,{
  .file <- str_remove(ppi.net.file, "[.]gz$")
  .url <- str_c(db.url, "/data/HI-union.tsv")
  download.file(.url, destfile = .file)
  R.utils::gzip(.file)
})
```

- Build a network from a list of interacting pairs

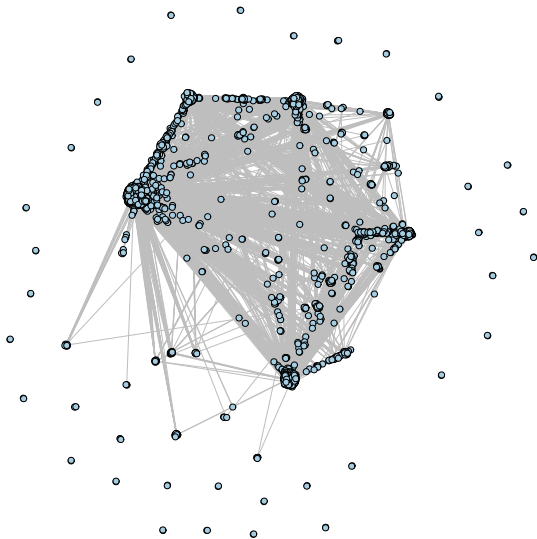
```
ppi.pairs <-
  fread(ppi.net.file, header=FALSE, col.names=c("v1", "v2")) %>%
  filter(v1 != v2) %>% as.matrix
G.ppi <- graph_from_edgelist(ppi.pairs, directed = FALSE)
```

- Vertices: genes (or the resulting proteins)
- Edges: two or multiple genes work together (could have many different meanings)

```
n <- length(V(G.ppi))
m <- length(E(G.ppi))
```

- $n = 9060$ vertices
- $m = 63242$ edges

Human Reference Protein-Protein interaction map



- Vertices: genes (or the resulting proteins)
 - Edges: two or multiple genes work together (could have many different meanings)
 - $n = 9060$ vertices
 - $m = 63242$ edges
- Sources of biological networks:
- High-throughput experiments
 - Co-expression analysis
 - Literature-based curation
- (Luck et al. 2020; Rual et al. 2005)

Degree distribution (How many friends?)

Vertex degree

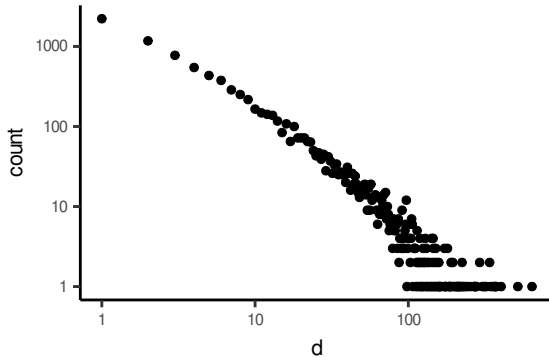
For each vertex $v \in V(G)$: $d_v = \sum_u A_{uv}$

- It is simple to calculate in igraph:
- Degree distributions may teach us the underlying generative scheme
- Power-law (almost linear slope in a log-log plot) may indicate a “rich-get-richer” phenomenon.
- The majority with < 10 neighbours vs. hub nodes with > 100 neighbours
- Poisson distribution if edges are drawn uniformly at random

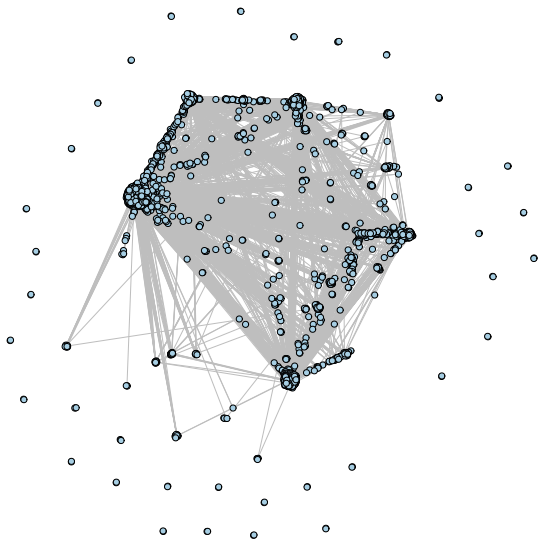
Degree distribution (How many friends?)

A handy one-liner:

Log-log plot:



Connected components and a giant component



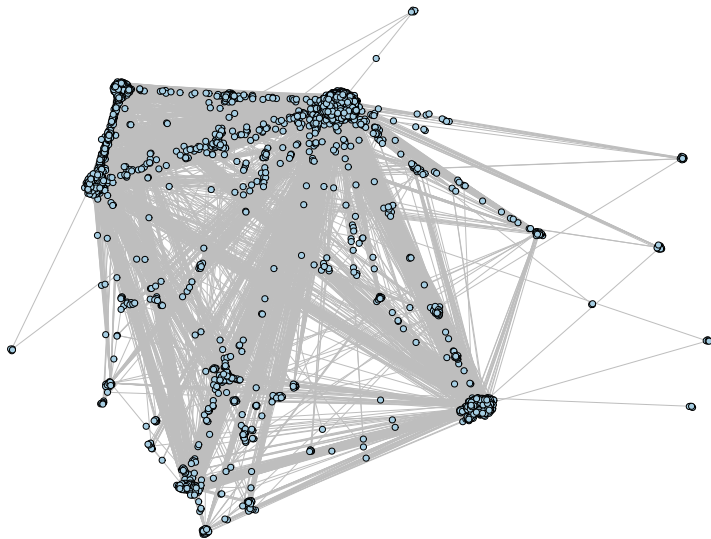
A (connected) component = a set of vertices reachable by hopping through edges.

Found 36 connected components

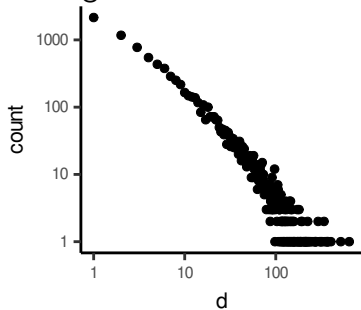
n	count
8986	1
2	31
3	4

- Can we take the giant one?

Only the giant component in PPI



- $n(G)$: 8986 (vs. 9060)
- $m(G)$: 63203 (vs. 63242)
- Degree distribution:

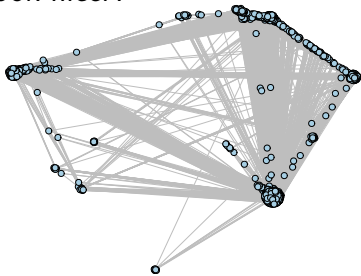


Iterative degree cutoff to remove dangling vertices

We may consider this as a QC step:
Comparison with the raw PPI data:

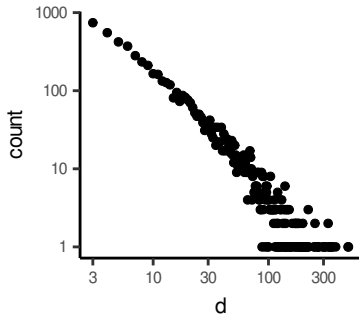
- $n(G)$: 5525 (vs. 9060)
- $m(G)$: 58548 (vs. 63242)

Will it look nicer?



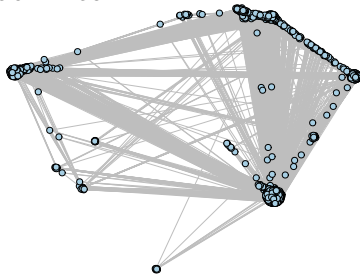
Iterative degree cutoff to remove dangling vertices

Degree distribution:



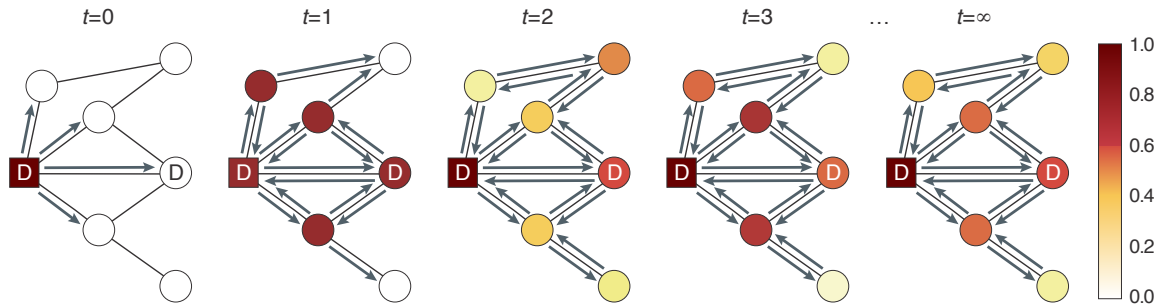
- $n(G)$: 5525 (vs. 9060)
- $m(G)$: 58548 (vs. 63242)
- Note: d starts from 3

Will it look nicer?



- Rationale: no genes act alone. If some genes have significantly fewer edges connected, high-throughput experiments may not have covered them sufficiently.

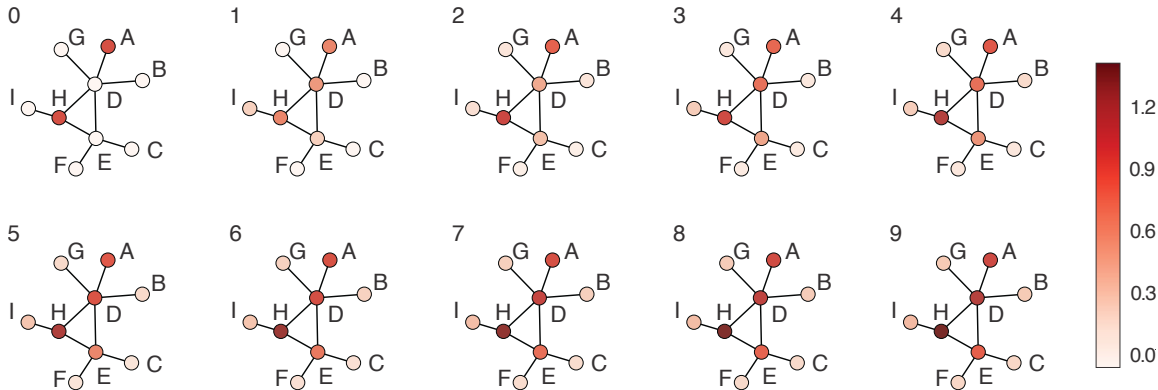
Random walks in network amplify associations



Square “D” is the original hit $\implies$ found circled “D”

(Cowen et al. 2017)

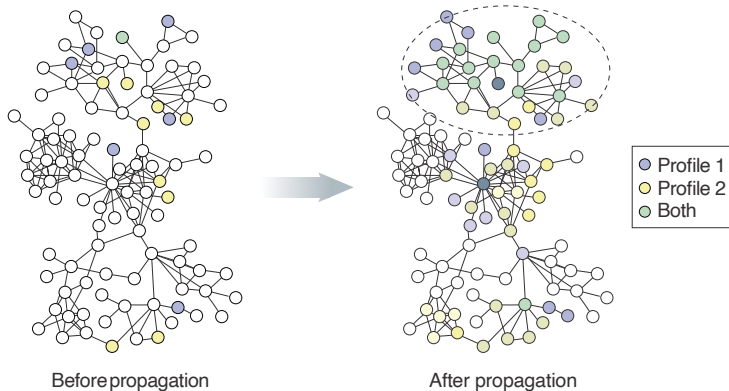
Propagation amplifies disease associations



Network propagation strengthens weak genetic signals by spreading information across interacting genes.

(Cowen et al. 2017)

Network Propagation as natural data integration



(Cowen et al. 2017)

- Different data types (mutations, expression, pathways) mapped onto network
- Network propagation combines signals across data sources
- Unified mathematical framework for genetic discovery

Example: Gene prioritisation in Multiple Sclerosis

Challenge:

- GWAS identifies ~100 MS-associated genes
- But these represent only a fraction of disease biology
- Many relevant genes may not show strong GWAS signal
- How to find additional disease-relevant genes?

Solution: Network propagation

- Start with MS GWAS genes as seeds
- Propagate signal through protein-protein interaction network
- Genes frequently “visited” by random walks are prioritised

Random walk on PPI network

Mathematical framework:

$$\mathbf{p}^{(t)} \leftarrow \gamma W \mathbf{p}^{(t-1)} + (1 - \gamma) \mathbf{p}^{(0)}$$

- $\mathbf{p}^{(0)}$: initial probability (1 for MS genes, 0 otherwise)
- W : transition matrix from adjacency matrix
- γ : restart probability (typically 0.85)
- Iterate until convergence

Result:

- Genes close to MS seeds in network receive high scores
- Captures functional relationships beyond direct interactions

Step 1: Download MS GWAS genes

- Download GWAS Catalog from NHGRI-EBI
- Extract genes associated with “Multiple Sclerosis”
- **Result:** 751 MS-associated genes

Step 2: Load human PPI network

- BioGRID human protein-protein interactions
- Uses gene symbols (not Ensembl IDs)
- **Result:** 20,448 genes, 1,239,391 interactions

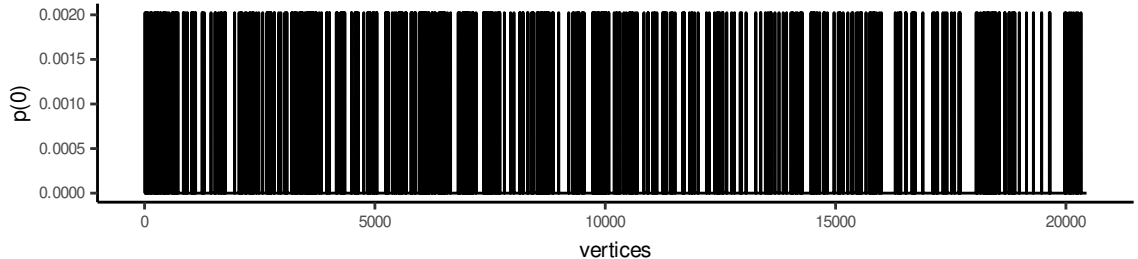
Step 3: Map MS genes to network

- BioGRID uses gene symbols - direct matching with MS genes
- **Result:** 494 of 751 MS genes found in network
- **2.4%** of network genes
- Need to expand beyond direct hits via propagation

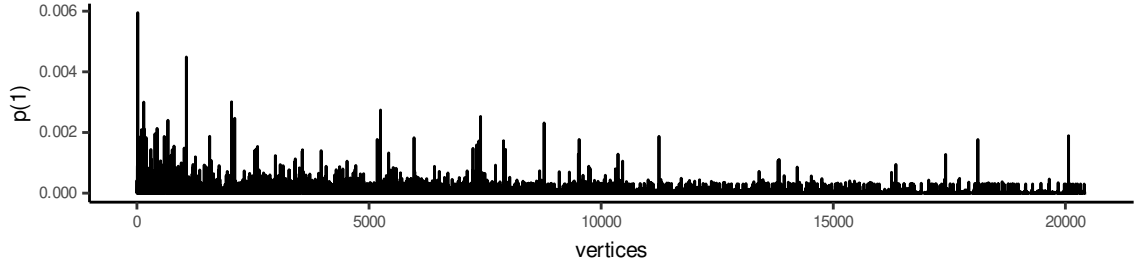
Step 4: Random walk with restart

- Initialize: MS genes = 1, others = 0
- Iterate: $\mathbf{p}^{(t)} \leftarrow 0.85 \times W\mathbf{p}^{(t-1)} + 0.15 \times \mathbf{p}^{(0)}$
- Run 100 iterations until convergence
- **Result:** All genes ranked by propagation score

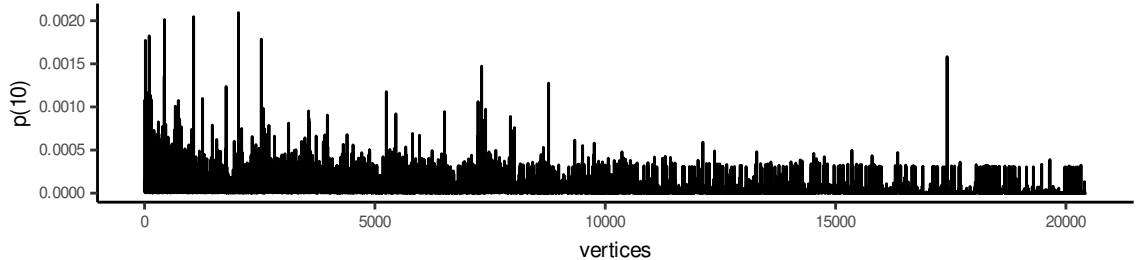
Probability spreads through network over iterations



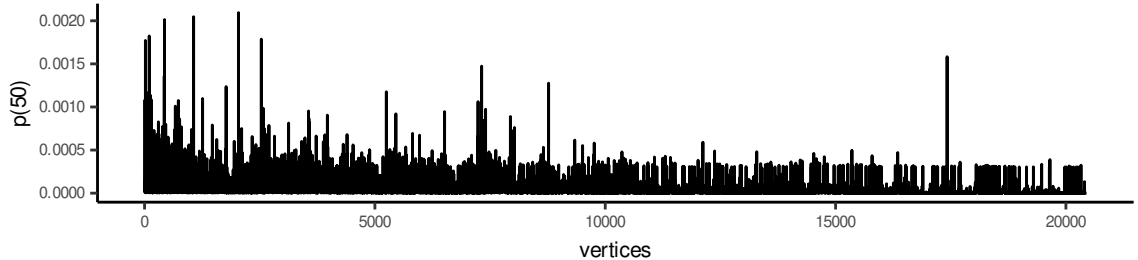
Probability spreads through network over iterations



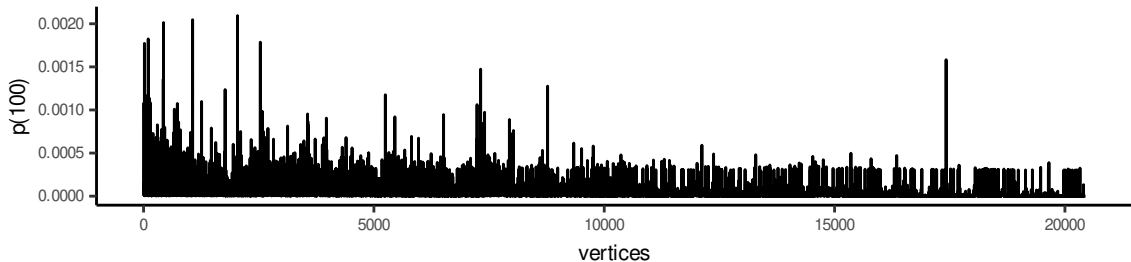
Probability spreads through network over iterations



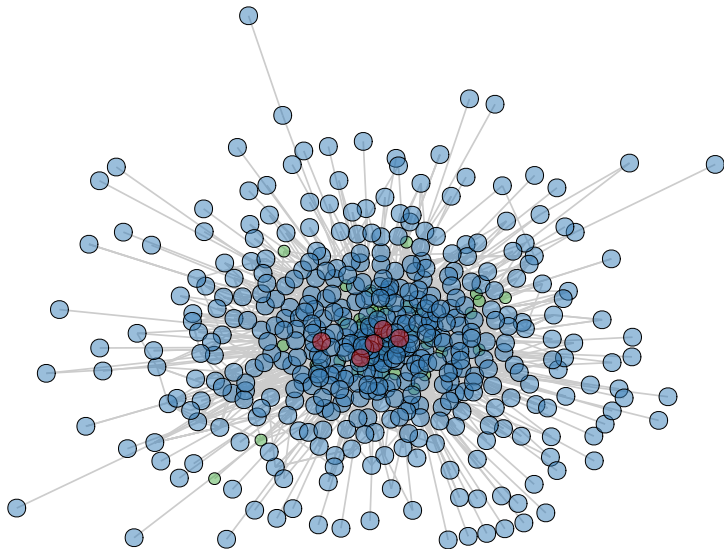
Probability spreads through network over iterations



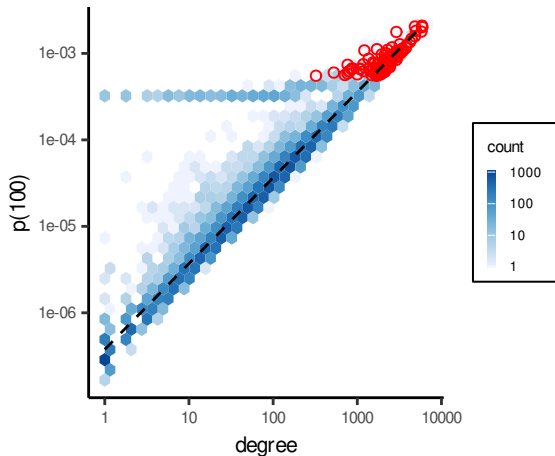
Probability spreads through network over iterations



Top 50 propagation genes + all MS genes in network



Are we just finding high-degree vertices?

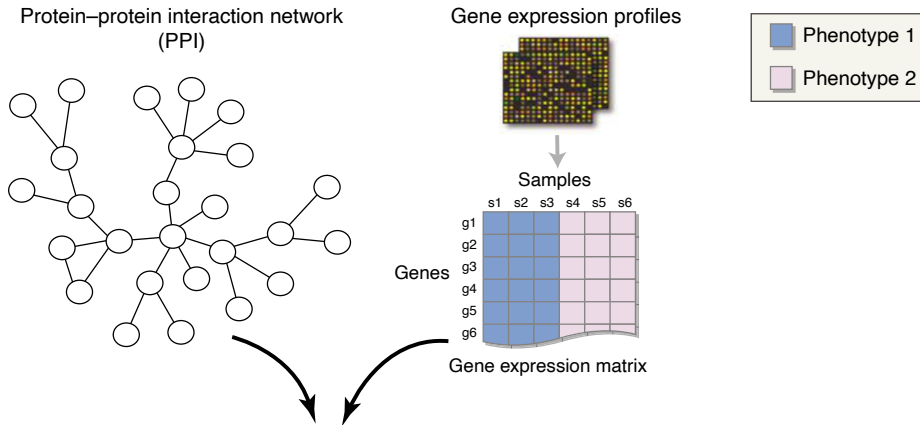


Red dots: top 100 genes

Network propagation summary

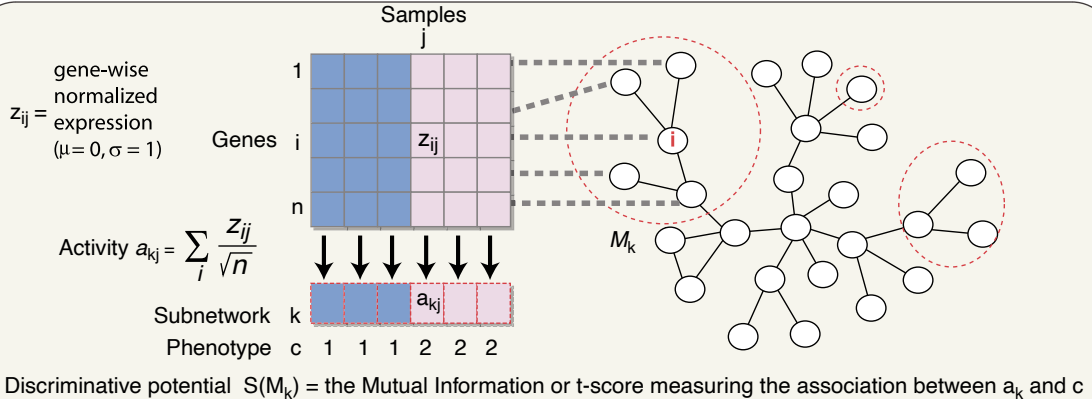
- Start with seed genes (e.g., from GWAS, mutations)
- Propagate signal through the network
- Neighbors of seed genes receive scores based on network distance
- Discover related genes not in original list
- Aggregates weak signals across pathways
- Applications: disease gene prioritization, drug target discovery

Integrate PPI with tumour expression profiles



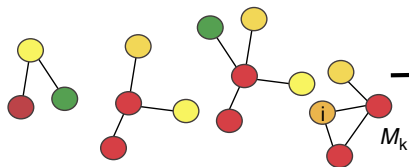
Chuang et al. (2007)

Subnetwork activity aggregates coordinated gene expression

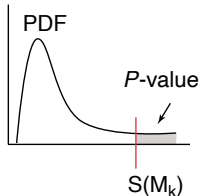


Chuang et al. (2007)

Search for subnetworks maximally discriminative of phenotype



Subnetworks maximizing $S(M_k)$
for each starting node in PPI



$p1$:

The null distribution of S is estimated
by all random subnetworks

$p2$:

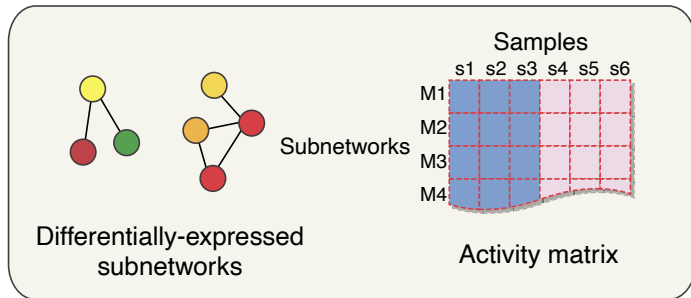
The null distribution of $S(M_k)$ is estimated
by random subnetworks seeded at node i

$p3$:

The null distribution of $S(M_k)$ is estimated
by permuting phenotypes

Chuang et al. (2007)

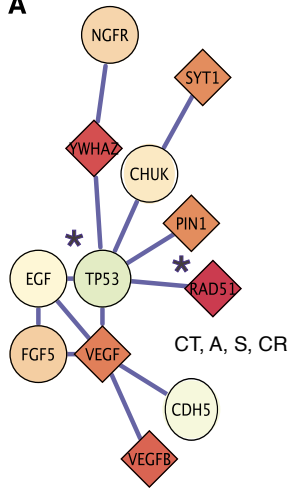
Significant subnetworks $\Rightarrow$ a feature matrix for classification of many relevant genes



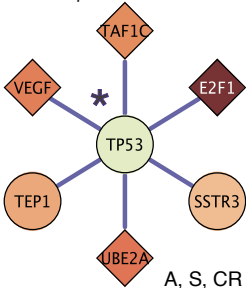
Chuang et al. (2007)

Subnetworks centre on TP53 and ERBB2 hubs

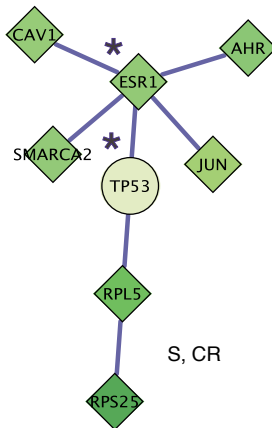
A



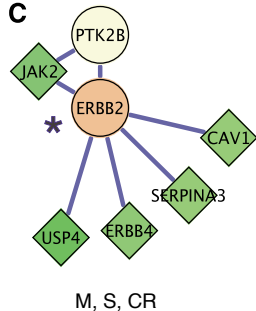
- (M) Metabolism
- (CT) Cell and tissue remodeling
- (CC) Circulation and coagulation
- (A) Apoptosis
- (S) Signaling of cell growth and survival
- (CR) Cell proliferation and replication



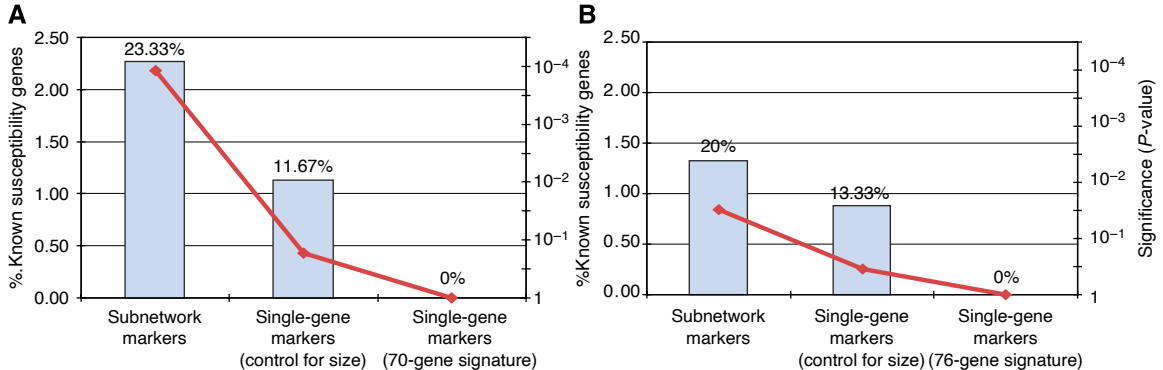
B



C



Subnetworks recover more known susceptibility genes than single-gene markers



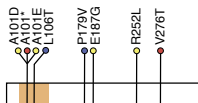
Chuang et al. (2007)

Most cancer genes are rarely mutated

Input mutation data

SNVs and small indels

293,863 SNVs in 20,473 genes



CNAs

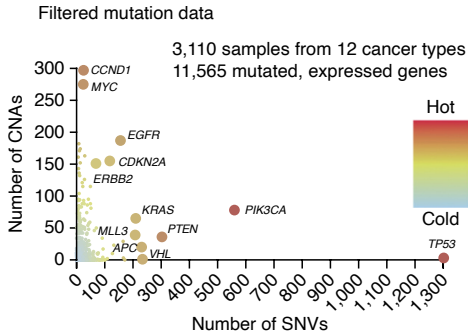
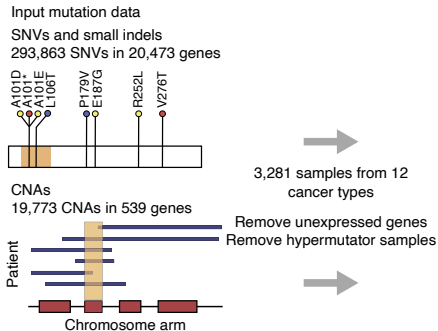
19,773 CNAs in 539 genes



3,281 samples from 12 cancer types

Leiserson et al. (2015)

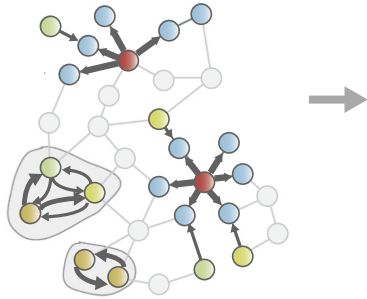
Most cancer genes are rarely mutated



Leiserson et al. (2015)

HotNet2: heat diffusion through PPI finds connected altered subnetworks

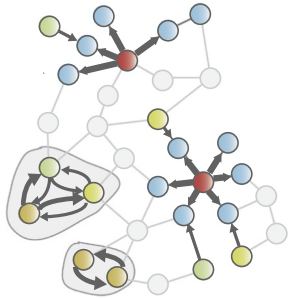
HotNet2 algorithm



Leiserson et al. (2015)

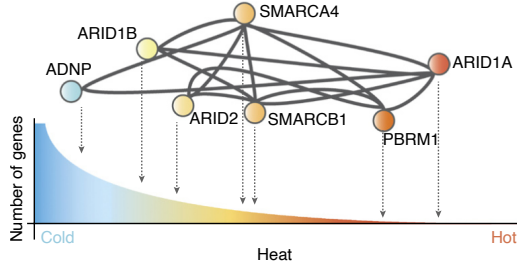
HotNet2: heat diffusion through PPI finds connected altered subnetworks

HotNet2 algorithm



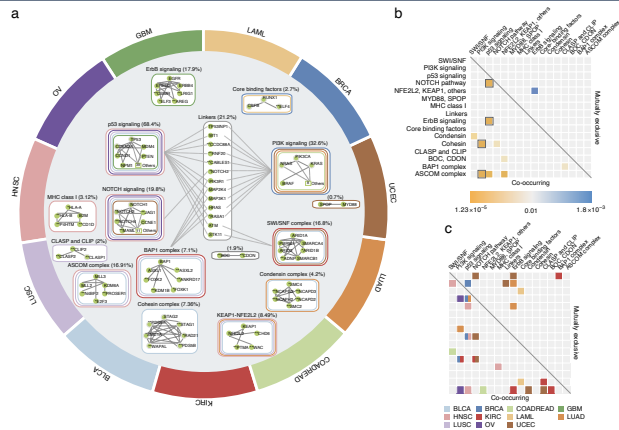
Significantly mutated subnetworks and complexes
16 subnetworks containing 147 genes

SWI/SNF complex



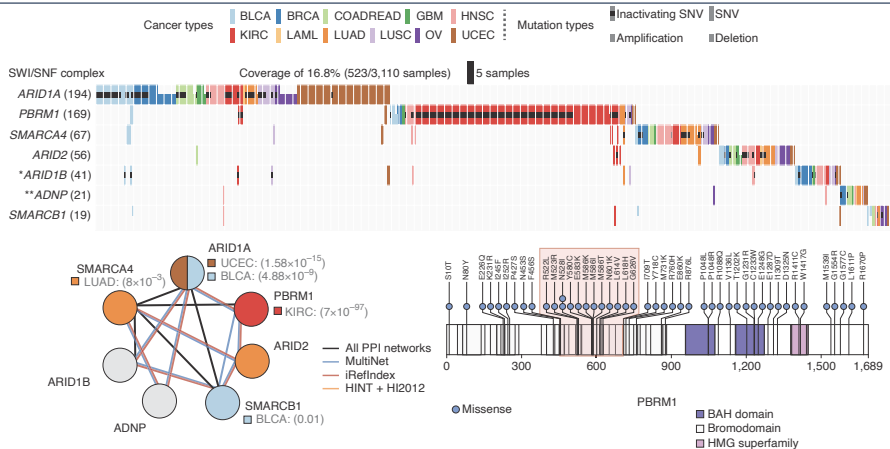
Leiserson et al. (2015)

Subnetworks map to known pathways and show co-occurrence patterns across cancer types



Leiserson et al. (2015)

SWI/SNF chromatin remodelling complex altered in 16.8% of pan-cancer samples



Leiserson et al. (2015)

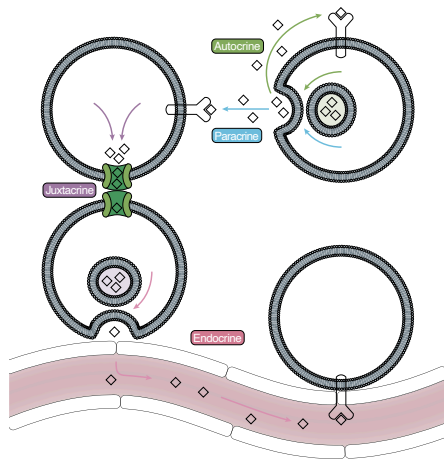
Pan-cancer network analysis

- Cancer is driven by pathways, not just individual genes
- Challenge: different patients have mutations in different genes
- Solution: project mutations onto PPI network, find mutated subnetworks
- HotNet2 identifies altered pathways and protein complexes
- Reveals cancer driver pathways missed by single-gene analysis

Today's lecture

- 1 How do we measure protein/gene interaction networks?
- 2 How can we use network data?
- 3 New frontiers: cell-cell communication and regulatory perturbation

Cells communicate with one another



Autocrine: cell signals to itself

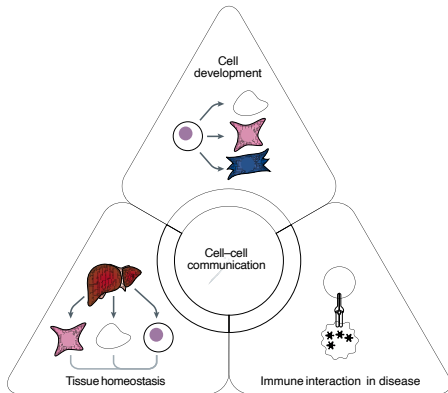
Paracrine: signals to nearby cells

Endocrine: signals via
bloodstream to distant cells

Juxtacrine: signals through direct
membrane contact

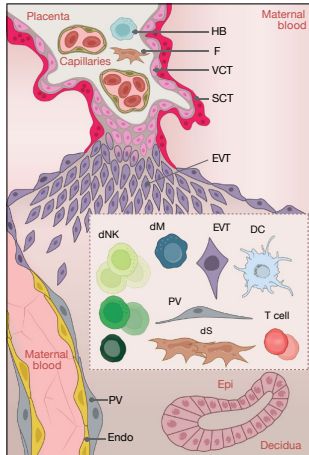
(Armingol et al. 2021)

Cell-cell communication shapes development, tissue homeostasis, and disease

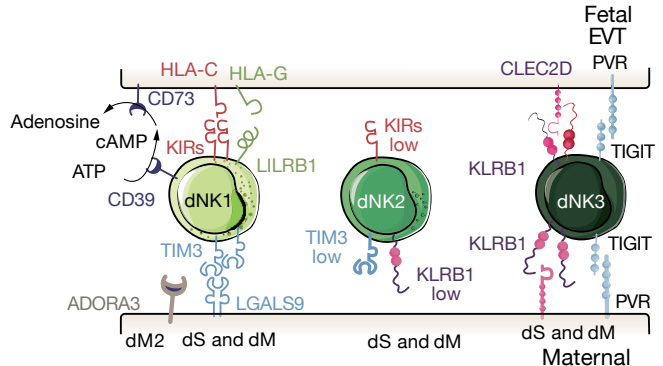


(Armingol et al. 2021)

Maternal-fetal interface $\longleftrightarrow$ cell-cell interactions



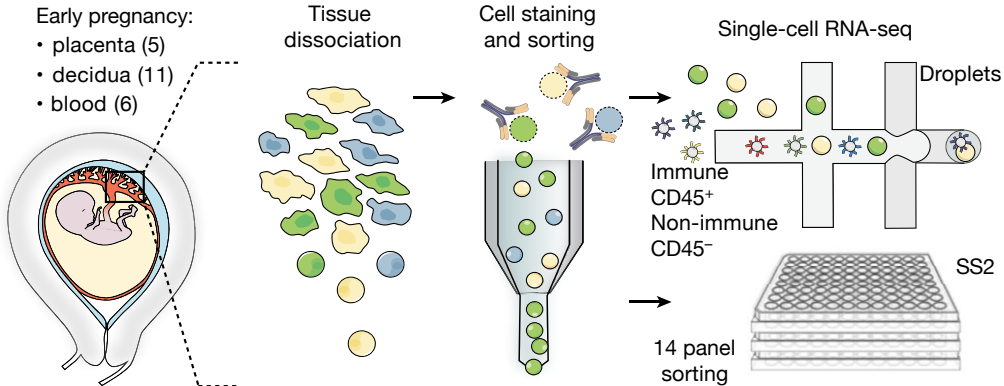
(Vento-Tormo et al. 2018)



Decidual^a NK cells coordinate trophoblast recruitment and tissue remodelling

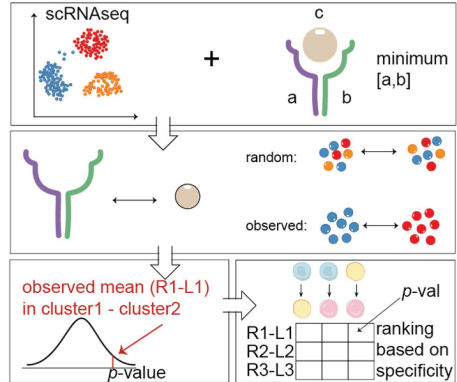
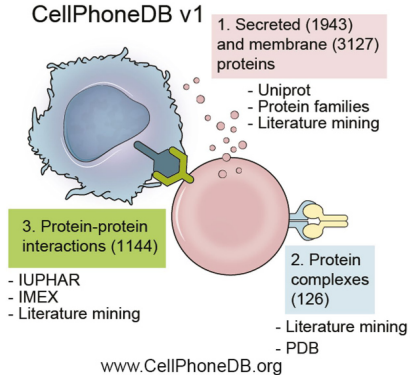
^apregnant uterine lining

Single-cell atlas reveals diverse cell types at the maternal-fetal interface



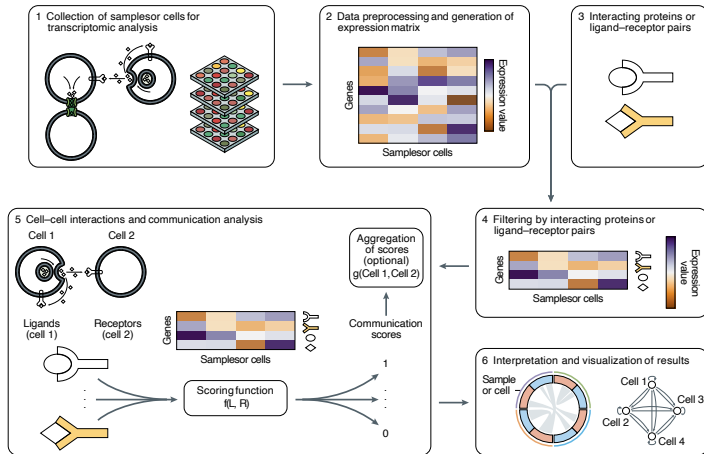
(Vento-Tormo et al. 2018)

CellPhoneDB scores ligand-receptor pairs using mean expression and permutation testing



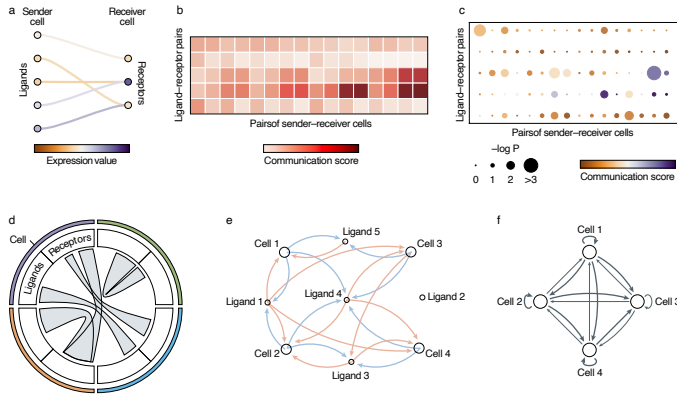
(Vento-Tormo et al. 2018)

Inferring cell-cell communication from scRNA-seq



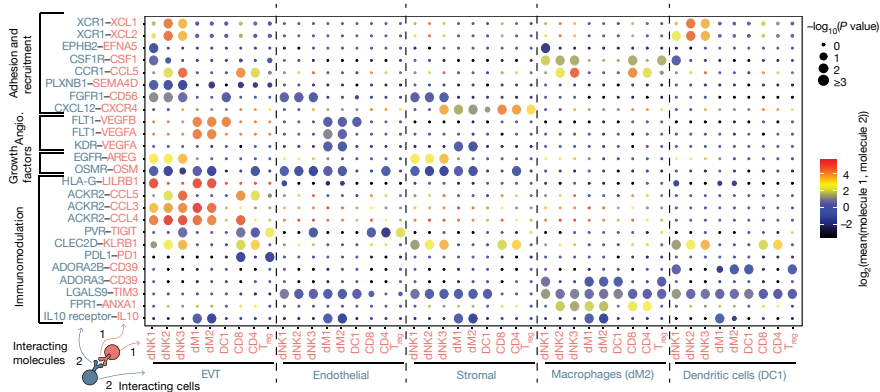
(Armingol et al. 2021)

Visualizing cell-cell communication scores across ligand-receptor pairs and cell types



(Armingol et al. 2021)

Decidual NK subsets modulate immune tolerance via HLA-C/KIR and checkpoint interactions



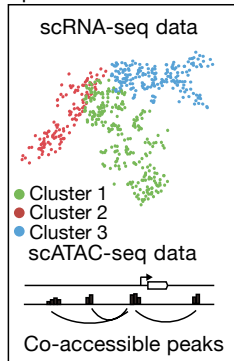
(Vento-Tormo et al. 2018)

Cell-cell communication in tissues

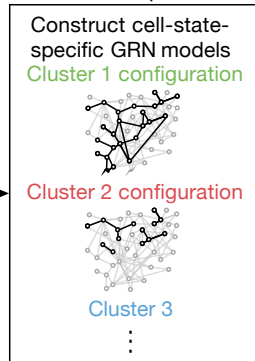
- Single-cell atlas of maternal-fetal interface (early pregnancy)
- Mapped ligand-receptor interactions between immune, stromal, and trophoblast cells
- Revealed communication networks maintaining pregnancy
- Showed how immune tolerance is established at the maternal-fetal boundary

CellOracle: from multi-omics to GRN to in silico TF perturbation

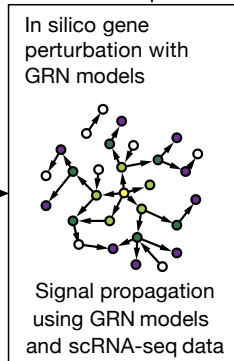
Input data



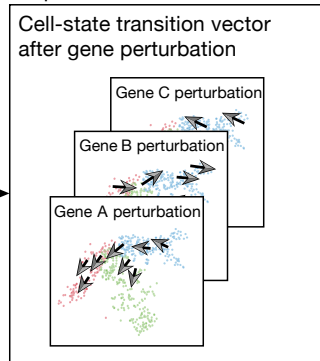
Calculation step 1



Calculation step 2



Output



(Kamimoto et al. 2023)

Chromatin accessibility defines the regulatory wiring

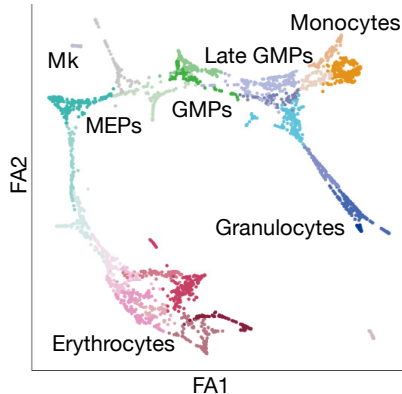
- scATAC-seq open chromatin defines $\mathcal{R}(j)$: which TFs can regulate gene j

$$\hat{x}_j = \sum_{i \in \mathcal{R}(j)} \beta_{ij} x_i$$

- KO of TF k : set $x_k = 0 \rightarrow$ propagates through GRN $\rightarrow$ genome-wide $\Delta \mathbf{x}$
- Cell fate shift: cosine similarity between $\Delta \mathbf{x}_i$ and $(\mathbf{x}_j - \mathbf{x}_i)$ for each neighbor j

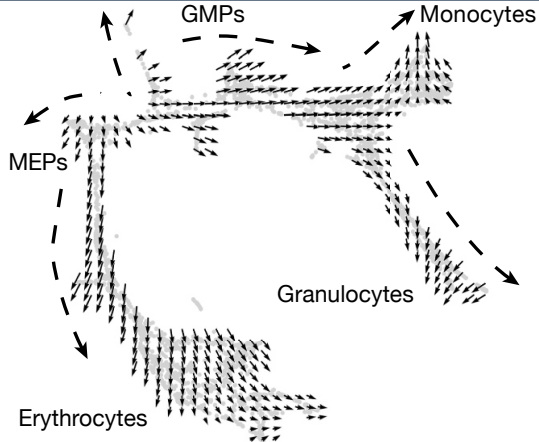
(Kamimoto et al. 2023)

Simulated differentiation flow recapitulates hematopoietic lineage trajectories



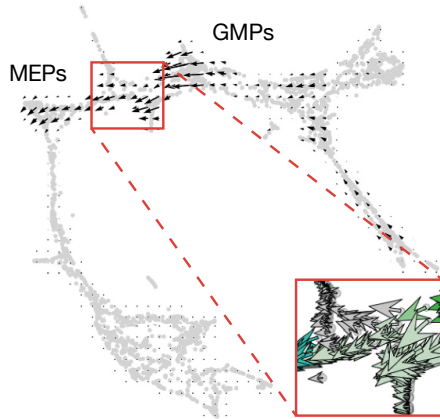
(Kamimoto et al. 2023)

In silico TF knockdown shifts cells toward erythroid/megakaryocyte fate



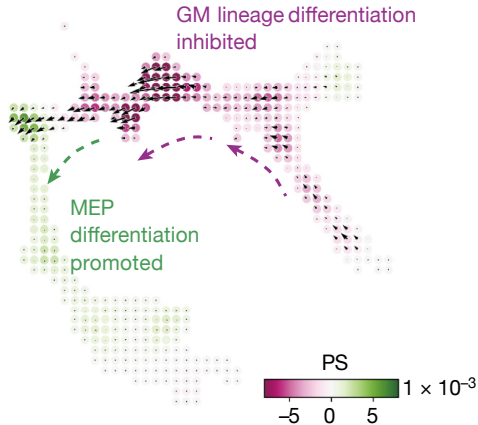
(Kamimoto et al. 2023)

Opposing TF knockdown promotes granulocyte-monocyte differentiation



(Kamimoto et al. 2023)

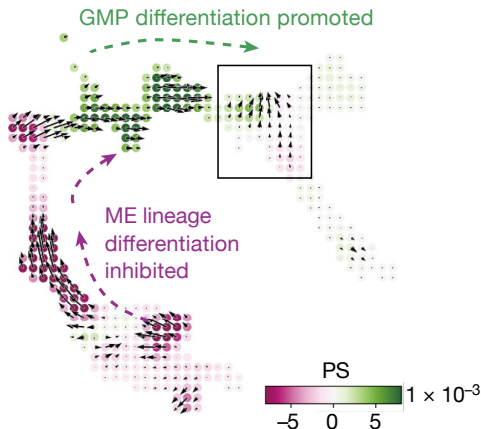
Spi1 KO: loss of a myeloid driver redirects cells toward erythroid fate



- Spi1 (PU.1) normally promotes GM/myeloid differentiation
- KO: GM lineage inhibited (magenta), MEP differentiation promoted (green)
- PS = perturbation score: signed measure of fate shift

(Kamimoto et al. 2023)

Gata1 KO has the opposite effect: ME inhibited, GMP promoted



- Gata1 normally promotes erythroid/ME differentiation
- KO: ME lineage inhibited (magenta), GMP differentiation promoted (green)
- Spi1 and Gata1 are mutual antagonists at the MEP/GMP branch point (Fig. 1g)

(Kamimoto et al. 2023)

Networks $\implies$ context for interpreting genomic data

Measuring networks:

- Physical interactions: Y2H, AP-MS
- Genetic interactions: synthetic lethality

Network analysis:

- Propagation: amplify signals across network
- Modules: disease genes cluster together

Cell-cell communication (CellPhoneDB):

- scRNA-seq + curated L-R database $\rightarrow$ who signals to whom

Intracellular GRN + perturbation (CellOracle):

- In silico TF KO $\rightarrow$ predict cell fate shift

References

- Adamic, Lada A, and Natalie Glance. 2005. "The Political Blogosphere and the 2004 US Election: Divided They Blog." In *Proceedings of the 3rd International Workshop on Link Discovery*, 36–43. ACM.
- Armingol, Erick, Adam Officer, Olivier Harismendy, and Nathan E Lewis. 2021. "Deciphering Cell-Cell Interactions and Communication from Gene Expression." *Nat. Rev. Genet.* 22 (2): 71–88.
- Chuang, Han-Yu, Eunjung Lee, Yu-Tsueng Liu, Doheon Lee, and Trey Ideker. 2007. "Network-Based Classification of Breast Cancer Metastasis." *Mol. Syst. Biol.* 3 (1): 140.
- Cowen, Lenore, Trey Ideker, Benjamin J Raphael, and Roded Sharan. 2017. "Network propagation: a universal amplifier of genetic associations." *Nat. Rev. Genet.* 18 (9): 551–62.
- Fields, S, and O Song. 1989. "A Novel Genetic System to Detect Protein-Protein Interactions." *Nature* 340 (6230): 245–46.

References

- Gavin, A C, M Bösche, R Krause, P Grandi, M Marzioch, A Bauer, J Schultz, et al. 2002. "Functional Organization of the Yeast Proteome by Systematic Analysis of Protein Complexes." *Nature* 415 (6868): 141–47.
- Girvan, M, and M E J Newman. 2002. "Community Structure in Social and Biological Networks." *Proc. Natl. Acad. Sci. USA* 99 (12): 7821–26.
- Kaelin, W G, Jr. 2005. "The Concept of Synthetic Lethality in the Context of Anticancer Therapy." *Nat. Rev. Cancer* 5 (9): 689–98.
- Kamimoto, Kenji, Blerta Stringa, Christy M Hoffmann, Kunal Jindal, Lilianna Solnica-Krezel, and Samantha A Morris. 2023. "Dissecting Cell Identity via Network Inference and in Silico Gene Perturbation." *Nature* 614 (7949): 742–51.
- Leiserson, Mark D M, Fabio Vandin, Hsin-Ta Wu, Jason R Dobson, Jonathan V Eldridge, Jacob L Thomas, Alexandra Papoutsaki, et al. 2015. "Pan-Cancer Network Analysis Identifies Combinations of Rare Somatic Mutations Across Pathways and Protein Complexes." *Nat. Genet.* 47 (2): 106–14.

References

- Luck, Katja, Dae-Kyum Kim, Luke Lambourne, Kerstin Lazarina, Azadeh Moez, Manon Provost, Marc Brehme, et al. 2020. "A Reference Map of the Human Binary Protein Interactome." *Nature* 580 (7803): 402–8.
- Newman, M E J. 2001. "The Structure of Scientific Collaboration Networks." *Proc. Natl. Acad. Sci. USA* 98 (2): 404–9.
- . 2006. "Finding Community Structure in Networks Using the Eigenvectors of Matrices." *Phys. Rev. E* 74 (3): 036104.
- Rual, Jean-François, Kavitha Venkatesan, Tong Hao, Tomoko Hirozane-Kishikawa, Amélie Dricot, Ning Li, Gabriel F Berriz, et al. 2005. "Towards a Proteome-Scale Map of the Human Protein-Protein Interaction Network." *Nature* 437 (7062): 1173–78.
- Tong, A H, M Evangelista, A B Parsons, H Xu, G D Bader, N Pagé, M Robinson, et al. 2001. "Systematic Genetic Analysis with Ordered Arrays of Yeast Deletion Mutants." *Science* 294 (5550): 2364–68.

References

- Uetz, P, L Giot, G Cagney, T A Mansfield, R S Judson, J R Knight, D Lockshon, et al. 2000. "A Comprehensive Analysis of Protein-Protein Interactions in *Saccharomyces Cerevisiae*." *Nature* 403 (6770): 623–27.
- Vento-Tormo, Roser, Mirjana Efremova, Rachel A Botting, Margherita Y Turco, Miquel Vento-Tormo, Kerstin B Meyer, Jong-Eun Park, et al. 2018. "Single-Cell Reconstruction of the Early Maternal-Fetal Interface in Humans." *Nature* 563 (7731): 347–53.
- Ye, Ping, Brian D Peyser, Xuewen Pan, Jef D Boeke, Forrest A Spencer, and Joel S Bader. 2005. "Gene Function Prediction from Congruent Synthetic Lethal Interactions in Yeast." *Mol. Syst. Biol.* 1 (1): 2005.0026.